Day 2 – Routers & Switches (Practical + Commands)

1. Check Network Adapter

Command

```
ipconfig
```

Purpose

- View the current IP Address.
- View the Default Gateway.
- Check whether the computer has received an IP address from DHCP.

Example

```
IPv4 Address . . . . . : 192.168.1.100
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . : 192.168.1.1
```

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2. View Detailed Network Information

Command

```
ipconfig /all
```

Displays

- MAC Address
- DHCP Status
- DNS Server
- Lease Time
- Default Gateway

Interview Question

What is the difference between `ipconfig` and `ipconfig /all`?

Answer

- `ipconfig` displays basic network information.
- `ipconfig /all` displays detailed network information.

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3. Refresh the IP Address

Sometimes the computer receives an incorrect IP address.

Release the Current IP

```
ipconfig /release
```

Request a New IP

```
ipconfig /renew
```

Use When

- No Internet connection
- Incorrect IP Address
- DHCP-related issues

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4. Clear the DNS Cache

Command

```
ipconfig /flushdns
```

Use

Clears old DNS records stored on the computer.

Useful when websites are not opening because of outdated DNS cache.

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5. Test the Router

Command

```
ping 192.168.1.1
```

Successful Output

```
Reply from 192.168.1.1
```

Failed Output

```
Request Timed Out
```

Meaning

The computer cannot communicate with the router.

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6. Test Internet Connectivity

Command

```
ping 8.8.8.8
```

Google Public DNS Server

If Successful

The Internet connection is working.

If Failed

There may be a problem with the ISP or the router.

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7. Test DNS Resolution

Command

```
ping google.com
```

If:

- `ping 8.8.8.8` works
- but `ping google.com` fails

Then the problem is related to the DNS server.

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8. Check Physical Network Connection

Command

```
netsh interface show interface
```

Displays

- Connected
- Disconnected
- Disabled

---

9. View MAC Address

Command

```
getmac
```

or

```
ipconfig /all
```

Example

```
40-8D-5C-7B-11-22
```

A switch uses the MAC Address to forward data.

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10. View the ARP Table

Command

```
arp -a
```

Displays

IP Address ↔ MAC Address Mapping

Example

```
192.168.1.1
00-11-22-AA-BB-CC
```

Interview Question

What is ARP?

Answer

ARP (Address Resolution Protocol) converts an IP Address into a MAC Address.

Switch Practical

Suppose you have the following setup:

```
PC1
  |
Switch
  |
PC2
```

Copy a file from one computer to another.

Observe the **Activity LED** on the switch.

If the LED blinks, it means data is being transferred successfully.

Open Network Connections

Command

```
ncpa.cpl
```

This command opens the Network Connections window.

Open Command Prompt Quickly

Press

`Win + R`

Type

`cmd`

Press Enter.

Router Practical

Common Default Gateway Addresses

`192.168.1.1`

`192.168.0.1`

`192.168.10.1`

Open a web browser and enter

`http://192.168.1.1`

The router login page should open.

Day 2 Interview Questions

Router works on which OSI Layer?

Layer 3 (Network Layer)

Switch works on which OSI Layer?

Layer 2 (Data Link Layer)

Router uses which address?

IP Address

Switch uses which address?

MAC Address

What does a Router connect?

Different Networks

What does a Switch connect?

Devices within the same network.

Which cable connects a Router to a Switch?

Ethernet Cable (RJ45)

What is the usual color of the WAN Port?

Usually Blue.

What is the usual color of LAN Ports?

Usually Yellow.

(Note: Colors may vary depending on the manufacturer.)

Troubleshooting Tips

Power LED OFF

Check:

- Power Adapter
- Power Switch
- Power Socket

LAN LED OFF

Check:

- Ethernet Cable
- LAN Port
- Network Interface Card (NIC)

Internet LED OFF

Check:

- WAN Cable
- ISP Connection

Ping Failed

Run

```
ipconfig
```

Then verify:

- Default Gateway
- Ethernet Cable
- Firewall Settings

Day 3 Practical Commands

View IP Address

```
ipconfig
```

View Detailed Network Information

```
ipconfig /all
```

Ping the Router

```
ping 192.168.1.1
```

Ping Google Public DNS

```
ping 8.8.8.8
```

Ping a Website

```
ping google.com
```

Continuous Ping

```
ping 192.168.1.1 -t
```

Purpose

Checks network stability continuously.

Press

`Ctrl + C`

to stop the test.

Trace the Route

```
tracert google.com
```

Displays every router between your computer and Google's server.

DNS Lookup

```
nslookup google.com
```

Displays:

- Website IP Address
- DNS Server

View Routing Table

```
route print
```

Displays the routing table of the computer.

View Network Statistics

```
netstat
```

Displays active network connections.

Detailed Version

```
netstat -ano
```

Displays

- Process ID (PID)
- Port Number
- Connection Status

View Shared Resources

```
net view
```

Displays shared computers and shared folders on the network.

View Computer Name

```
hostname
```

Displays the computer's hostname.

View System Information

```
systeminfo
```

Displays

- Operating System
- RAM
- BIOS
- Network Information

Router Configuration Tips

Always connect the computer to a **LAN Port**, **not** the WAN Port.

```
Computer
      |
 LAN Port
```

If the computer does not receive an IP Address,

check whether **DHCP is enabled**.

If a Static IP Address is required, configure:

```
IP Address : 192.168.1.20

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.1

DNS Server : 8.8.8.8
```

Practical Checklist

✔ Router Power LED ON

✔ LAN LED ON

✔ WAN Connected

✔ IP Address Assigned

✔ Default Gateway Available

✔ Router Responds to Ping

✔ Router Login Page Opens

✔ Internet Connection Working

Common Viva Questions

Question	Answer
What is DHCP?	DHCP automatically assigns IP Address, Subnet Mask, Default Gateway, and DNS Server.
What is a Default Gateway?	It is the device (usually the router) that connects the local network to other networks or the Internet.
Which command displays the Default Gateway?	`ipconfig`
Which command displays the MAC Address?	`getmac` or `ipconfig /all`
Which command checks DNS information?	`nslookup`
Which command displays the Routing Table?	`route print`

Question	Answer
Which command tests network connectivity?	`ping`
Which command traces the network path?	`tracert`
Which command clears the DNS cache?	`ipconfig /flushdns`
Which command releases the current IP Address?	`ipconfig /release`
Which command requests a new IP Address?	`ipconfig /renew`
Which command displays active network connections?	`netstat -ano`

Pro Tips (Real-World)

- Always connect the **Router LAN Port** to **any normal port on the Switch**. Most unmanaged switches do not require a dedicated uplink port.

- Run **Command Prompt as Administrator** because some networking commands require administrative privileges.
- If your computer receives an IP address like **169.254.x.x**, it usually means the DHCP server did not assign an IP address. Check the Ethernet cable, switch port, router, and DHCP service.
- Avoid creating a **network loop** by connecting two cables between the same router and switch or between two unmanaged switches. This can cause a **broadcast storm**, making the network very slow or completely unavailable.
- Follow this troubleshooting sequence whenever a network problem occurs:

```
Power
   ↓
Check Ethernet Cable
   ↓
Check LED Indicators
   ↓
Run ipconfig
   ↓
Ping the Default Gateway
   ↓
Ping 8.8.8.8
   ↓
```

```
Ping google.com
```

- To restore a router to its factory settings, press and hold the **Reset Button** for approximately **10–15 seconds**. (The exact time may vary depending on the router manufacturer.)

These practical commands, troubleshooting methods, and real-world tips are highly useful for **CCNA**, **Networking Support Engineer**, **Desktop Support Engineer**, **Hardware & Networking Technician**, and networking job interviews.